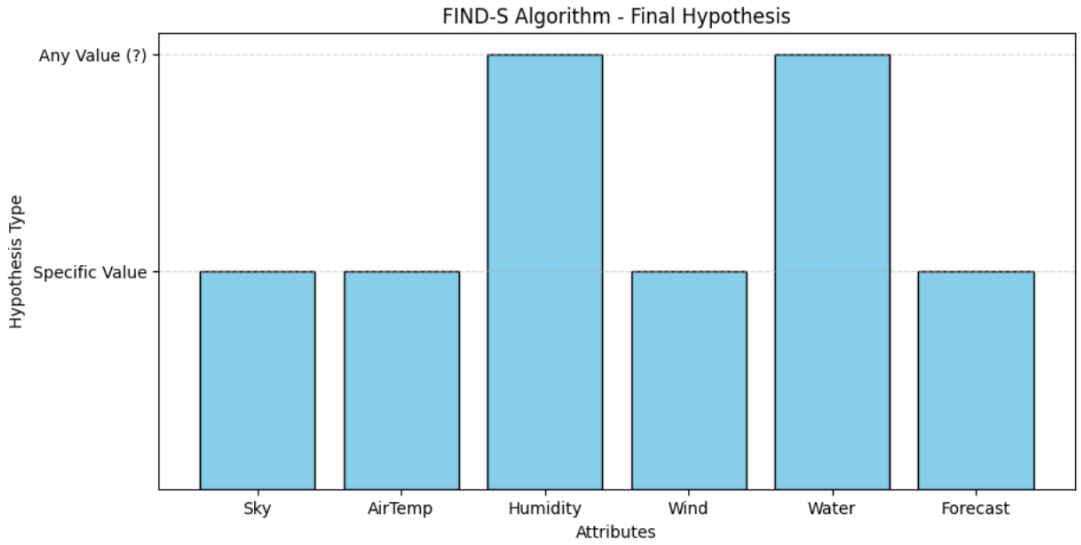
EXP 1 OUTPUT :

```
Training Dataset:
     Sky AirTemp Humidity    Wind Water Forecast EnjoySport
0  Sunny    Warm   Normal  Strong  Warm     Same        Yes
1  Sunny    Warm     High  Strong  Warm     Same        Yes
2  Rainy    Cold     High  Strong  Warm   Change         No
3  Sunny    Warm     High  Strong  Cool     Same        Yes

Steps of FIND-S Algorithm:
After positive example 1: ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']
After positive example 2: ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']
After positive example 4: ['Sunny', 'Warm', '?', 'Strong', '?', 'Same']

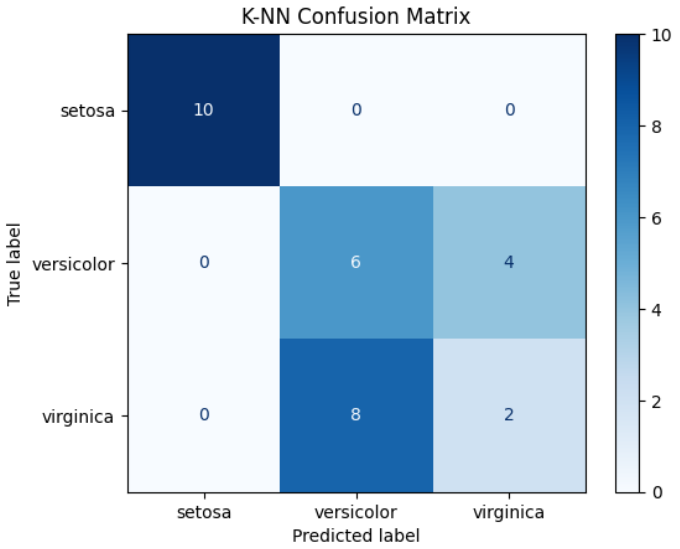
Final Most Specific Hypothesis:
['Sunny', 'Warm', '?', 'Strong', '?', 'Same']
```

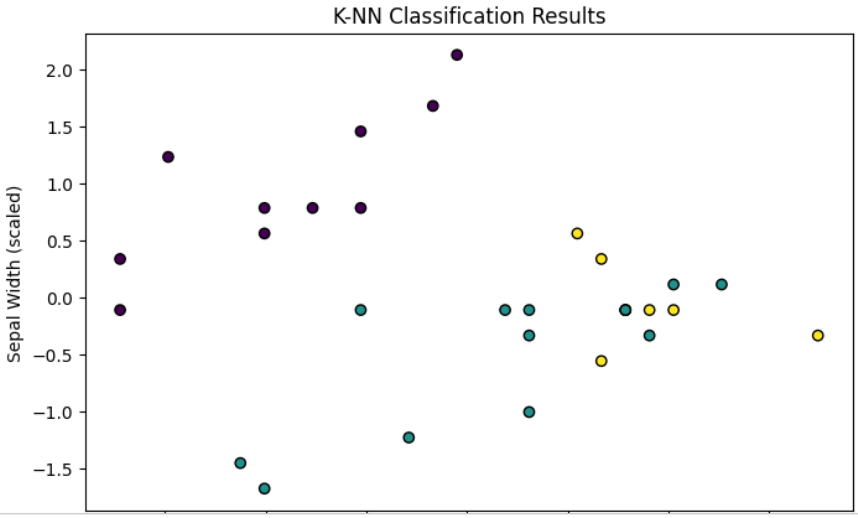


EXP 2 OUTPUT :

```
K-Nearest Neighbours Algorithm
Number of training samples: 120
Number of testing samples: 30
Accuracy: 60.0 %

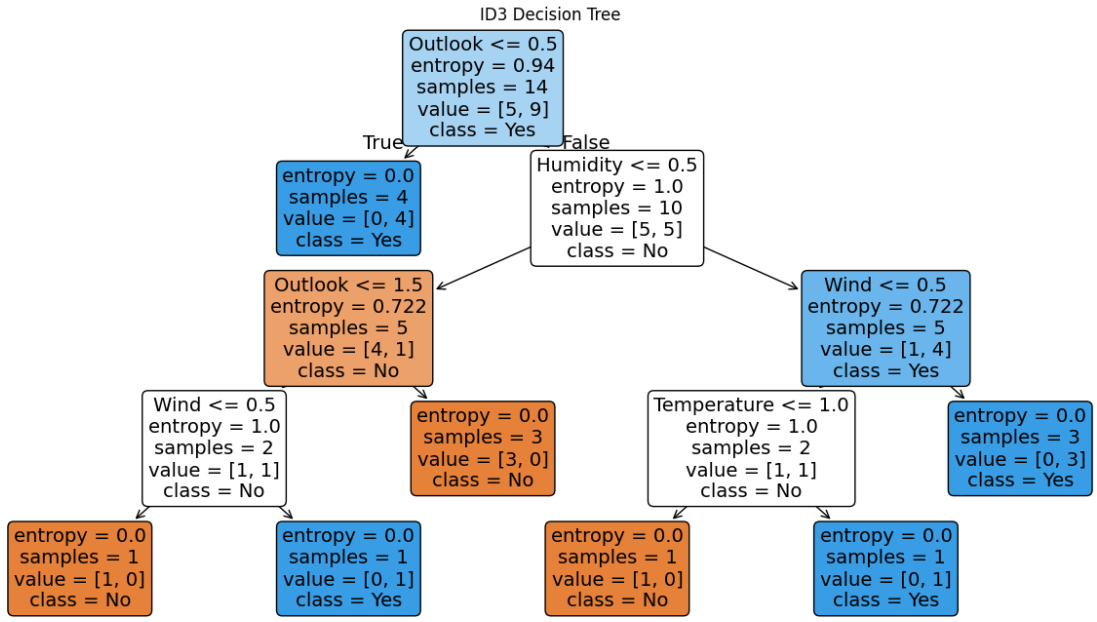
New Sample: [5.1, 3.5]
Predicted Class: setosa
```





EXP 3 OUTPUT:

```
Training Dataset:
     Outlook Temperature Humidity    Wind Play
0      Sunny         Hot     High    Weak   No
1      Sunny         Hot     High  Strong   No
2   Overcast         Hot     High    Weak  Yes
3       Rain        Mild     High    Weak  Yes
4       Rain        Cool   Normal    Weak  Yes
5       Rain        Cool   Normal  Strong   No
6   Overcast        Cool   Normal  Strong  Yes
7      Sunny        Mild     High    Weak   No
8      Sunny        Cool   Normal    Weak  Yes
9       Rain        Mild   Normal    Weak  Yes
10     Sunny        Mild   Normal  Strong  Yes
11  Overcast        Mild     High  Strong  Yes
12  Overcast         Hot   Normal    Weak  Yes
13      Rain        Mild     High  Strong   No
```




```
New Sample: {'Outlook': 'Sunny', 'Temperature': 'Cool', 'Humidity': 'Normal', 'Wind': 'Weak'}
Predicted Class: Yes
```

EXP 4 OUTPUT:

```
Artificial Neural Network using Backpropagation
Number of training samples: 120
 of testing samples: 30
Test Accuracy: 80.0 %
```

